The Comedy Cadence - The Math behind the Movie Magic

Analysis Report

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1. INTRODUCTION

Cinema is the most popular and powerful source of entertainment that the world has ever witnessed till date. In the tenure of this short and busy life, Cinema serves as the ultimate stressbuster. The intensity of impact created by the cinema is way more than that of a book or a newspaper. From guaranteeing emotional well-being to influencing the culture and perceptions, the power of the cinema extends on a large scale basis. But the influence of cinema is truly subjective. For some people, cinema is all about finding relaxation, for others, it is about learning, and so on.

Considering the love and passion for cinema, this project aims at integrating the ideal cinematic technical aspects with the application of data science. The major focus of this project relies on the 'Comedy' movies from the 'Hollywood' industry, for a 50 year time period (1977-2024).

Hollywood movies have marked their presence worldwide. Many great movies that had originated in this industry serve as an inspiration to the major blockbusters, industrial hits, and superhits in the local scale industries across the world. Understanding the technology behind creating and rendering humor to influence the huge mass of audience worldwide, especially in this context, would help this project in gaining a global touch and significance amongst the existing cinema technicians and the future cinema aspirants.

Humor is one such emotion, that is highly complexive, as well as highly demanded, all at the same time. Laughter is always an impactful and everlastingly appreciated reaction. In the world of cinema, making a movie based on the 'Comedy' genre is truly a matter of challenge, especially when it comes to the writing and the editing part. Getting to know the technique behind creating an impactful comedy and influencing the audience, with the help of data-oriented analysis is the basic aim of this project. To make the study more stronger and interesting, movies featuring unique humor types (14 in number) have been considered.

A full length movie is the resultant blend of camera angles, cuts, dialogue deliveries, and musical buildups. To mark the status of intensity for a scene, such techniques are essentially

needed to be balanced. Moreover, the impact of a scene is always expected to match with the satisfaction of the audience. When it comes to comedy, the hardest genre for any filmmaker to attempt and touch the glory of success, it is very much crucial to know the mechanism of integrating these technologies in framing a sequence. Ultimately, people must laugh, laugh out loud. In this context, this project majorly focuses on only the editing (Cuts) and script writing (Dialogue) aspects, to understand the so-called technically-related patterns on a baseline mode.

This project can sound beneficial to the lead technicians, including:

Director - The creative head, supervising every aspects of a movie

Screen Writer - The creative verbal genius, penning the screenplay for a movie

Cinematographer - The creative visual genius, fetching the beauty of the frames for a movie

Editor - The creative pacing genius, structuring the visuals for an impactful movie output

On an overall note, understanding the cinematic technology on a era-mathematical combination basis can help the technicians to determine and stage an ultimate cinematic experience, especially when it comes to the most demanded and highly appreciated comedy genre. An idea with respect to the balance or the proportion of verbal-visual aspects (Edits and Dialogues) in building up a sequence, can help the technicians to get a prior basic knowledge of under which category of humor-specific tension would the audiences' feeling fall in.

2. PROBLEM STATEMENT

Cinema is an absolute matter of subjective creativity. The uniqueness of any technician, be it a director, an editor, or a cinematographer, is being measured in terms of the level of creativity that prevails within. While the success of a movie praises the efforts put behind the execution of such creativity, the presence of technology adopted for such an execution is not much felt under the limelight.

For a complexive genre like comedy and to ensure maximum humor within a sequence, timing is an important factor to be considered. As a particular humor sequence guarantees laughter through the magic of timing, an idea on the mathematical mechanism adopted to guarantee this success, especially through bringing a balance between the editing (Cuts) and writing (Words) areas, is a lacking matter amongst the technicians.

The success of a movie demands a strong and unbreakable pattern of audience engagement. Getting to understand the effect of camera lingering (Static shot) in optimizing social discomfort, or in cinematic terms the 'cringe factor', helps the technicians like editors to balance their skill of cutting or prolonging a shot or a sequence, without compromising on the audience engagement. The actual metric, aiding in understanding the so-called cringe factor, is a matter for which the idea is lacking behind, and needs to be explored.

A successful movie also depends on the pace of its progression. From the beginning till the end of a movie, keeping up the energy level of the audience intact requires adopting an effective pacing mechanism on-screen. Such an elevation, from a simple introduction sequence to a highly-satisfying climax sequence, can be determined through a precise pacing mechanism, for which, a standardised idea is still lacking amongst the technicians.

As the era progresses, the expectations and preferences of the respective audience also changes. Though the style of the story and the humor can all greatly influence such expectations, getting to know about the mathematical mechanism of pacing the movie, especially the editing task, demands a great deal of attention. For this, understanding the era-specific boredom threshold is very much important, which is still not under the focus of cinema-related technology.

3. ASSUMPTIONS

This project is majorly driven upon the following assumptions:

1. Though Cinema is a subjective kind of entertainment, a humor-based cinema is a matter of enjoyment to all, irrespective of the era or the type of humor.

2. The major scenes considered for all movies (Hook, Conflict, Climax) are subjective, hence the duration, total cuts, word counts, and all varies accordingly. This implies that the outliers are a matter of exemption in this aspect.

4. OBJECTIVES

As the problem statements denote those aspects that are not imaginary and truly exists within the cinema industry, which is managed by the technicians mostly out of their creative-oriented gut feeling, this project majorly aims:

1. To understand the trajectory of Cuts and Dialogue mechanisms used in the era-specific comedy movies.

2. To determine the correlation between Cringe (Static prolonged shot) and Cut (Constantly edited shot), and understand the mechanism of implementing cuts and dialogues mathematically.

3. To understand the pacing style of the modern comedy movies, and frame a mathematically special Climax Acceleration Blueprint to design the movie progression pace accordingly.

4. To understand the era-specific Rhythm style, specially focusing on the Editing task, followed for each comedy movie, in order to frame an ideal benchmark useful for the editors.

5. To understand the statistical significance of the variables, and develop the cinematic tension framework by determining the ideal editing and dialogue numbers.

6. To understand the pattern of the technical metrics considered for Editing and Dialogues (Cuts per Minute and Words per Second), and predict the ideal tension categories.

5. DATASET DESCRIPTION

Methodology of data collection:

This project had aimed at an industry-oriented (Cinema industry), genre-specific (Comedy genre), global-level analytical approach. Hence, keeping the Hollywood cinema industry in mind (due to the worldwide presence), 15 comedy movies featuring various humor categories were chosen. The timeframe was considered for nearly 50 years (1977-2024), aiming at an era-specific level of understanding the technology-driven world cinema.

The data collection followed a track of focusing only on three major scenes for a specific movie that is considered crucial as well as aligns with the progression, namely the Hook scene, the Conflict scene, and the Climax scene.

The data collection was conducted from scratch, aiming to build a master data set with useful features and records that aligned with the objectives. For this, each movie was completely watched from the reliable OTT platforms (Amazon Prime, Netflix, and so on), while the data majorly focusing on the cinema-related technologies (Duration, Cuts, Word Counts) were all simultaneously collected using the Mobile device (Timer and Lap technique) on the basis of scene types, under the light of a movie. For better clarity on the information of the movies, Movie Title, Released Year, Humor type were all collected from the external sources (Google).

Within the timespan of 30 days, a complete master data set was framed and stored in the Google Sheets (Post which, imported as a CSV file in the Microsoft Excel) to proceed with the analysis further.

Feature-wise description:

Dataset name: Hollywood Cinema Data

Number of features: 9

Number of records: 45

1. Movie Title – Name of the movie (*String*)
2. Released Year – Year at which the movie was released (*Numeric*)
3. Humor Type – Type of the humor seen in a movie (*String*)
4. Scene Type – Scenes considered for the study (Hook, Conflict, Climax) (*String*)
5. Duration Sec – Duration of the scene (in seconds) (*Numeric*)
6. Total Cuts – Total number of cuts noted in a scene (*Numeric*)
7. Word Count – Total number of words within the dialogues in a scene (*Numeric*)
8. Tension Score – Humor specific tension score for the respective scene types (1-10) (*Numeric*)
9. Notes – Short description of the respective scene types (*String*)

6. ANALYSIS WORKFLOW

The analysis workflow for this project follows a particular pathway. From understanding the past pattern of using edits-dialogues in conveying the humor, till predicting the level of impact an audience would feel after experiencing the technically-tailored humor, the transition of raw data into a highly useful matter of decision making, found an evolution through the following steps:

Exploratory Data Analysis (EDA)
(Knowing the past)

⇩

Statistical Modelling
(Determining the statistical significance)
⇩
Prediction using Machine Learning
(Knowing the future)
⇩
Dashboard Analysis
(Trend in simple format)

Figure 6.1: Analysis workflow chart

6.1. EXPLORATORY DATA ANALYSIS (EDA)

Tool used: Python (Google Collab)

Link: [MDA_Project_EDA.ipynb]

Purpose:

To understand the pattern of pacing style and the level of sync between the technical variables (measures on Cuts and Dialogues) on multiple bases, relying upon the pre-defined objectives.

Coding workflow:

Importing the necessary libraries
(NumPy, Pandas, Matplotlib, Seaborn)
⇩
Loading the data set
(CSV format)
⇩
Conducting a preliminary check-up of the data
(Understanding the architecture using the relevant functions)
⇩
Objective 01: Visual vs. Verbal Dominance
(Creating a function for a granular-level of understanding, calculating additional measures if necessary, and plotting the chart)
⇩
Objective 02: Cringe-to-Cut correlation

(Creating a function for an aggregate-level of understanding, calculating additional measures if necessary, and plotting the chart)

⇩

Objective 03: The Climax Acceleration Blueprint

(Creating a function for a granular-level of understanding, calculating additional measures if necessary, and plotting the chart)

⇩

Objective 04: Era-specific Rhythm

(Creating a function for an aggregate-level of understanding, calculating additional measures if necessary, and plotting the chart)

⇩

Interpreting the insights

Figure 6.1.1: Exploratory Data Analysis - workflow chart

Outputs:

Objective 01: Visual vs. Verbal Dominance -

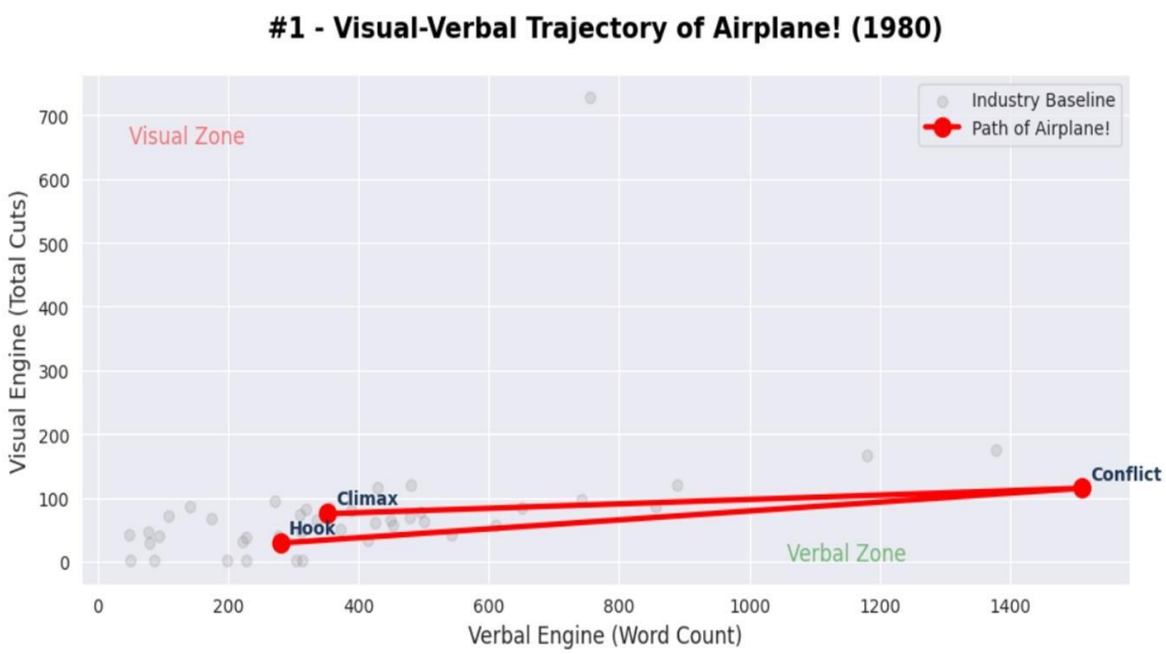

Figure 6.1.2: The Trajectory Map (Sample shown for the movie - Airplane! (1980))

Interpretation:

Movies find energy based on the intensity of words within the dialogue or the cuts on behalf of the editing mechanism. In cinema, technicians are required to understand the dynamics of factors that can enhance the experience that the audience can feel from watching a sequence. Here, each movie, representing unique types of humor, are being analyzed upon the trajectory or the progress, specially on the basis of the level of edits and dialogues combined together. While the pre-millennium era based movies (like the movie - *Airplane!* shown above), humor is highly relied on the cuts rather than the words (dialogue density is high). While in the case of post-millennium movies, the pattern depicts that, editing techniques are highly used to convey the humor, rather than the words (cuts are highly considered) in a sequence. This shows an era-based distinction in how the verbal engine and visual engine dominate with each other.

Objective 02: Cringe-to-Cut correlation -

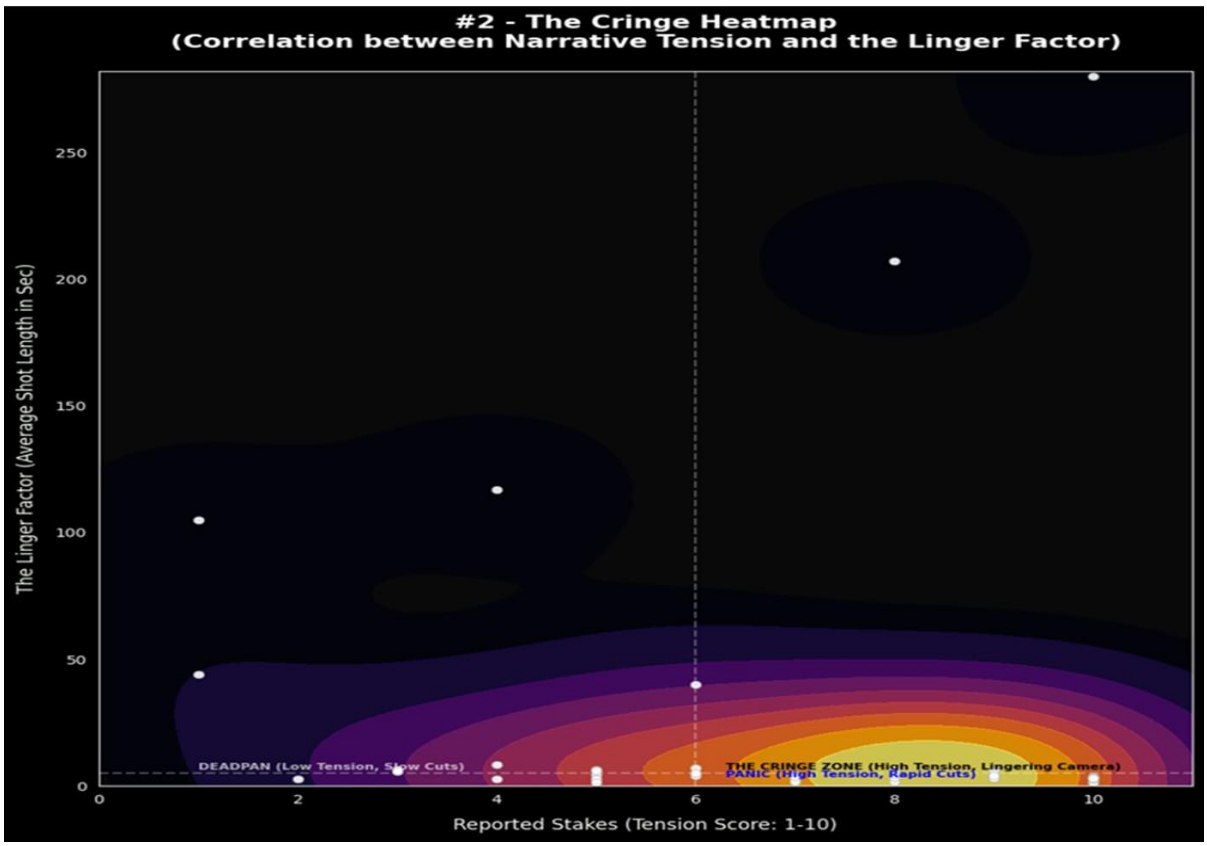

Figure 6.1.3: The Cringe Heatmap

Interpretation:

In cinema, the real cringe happens when a camera refuses to look away during a tense moment. As the shots are lingering, the level of tension that the audience should feel must be

higher. This means that, higher the Average shot Length (ASL), more should be the tension, neglecting the editing cuts to be an interruption. While most high-stakes scenes use fast "Panic" cuts, the outliers at the top of the chart hold a single shot for over 250 seconds. By lingering this long, the filmmaker forces the audience to sit through the awkwardness without giving a visual break. In this aspect, the power of silence or sharpness in the dialogues can be greatly felt.

Objective 03: The Climax Acceleration Blueprint -

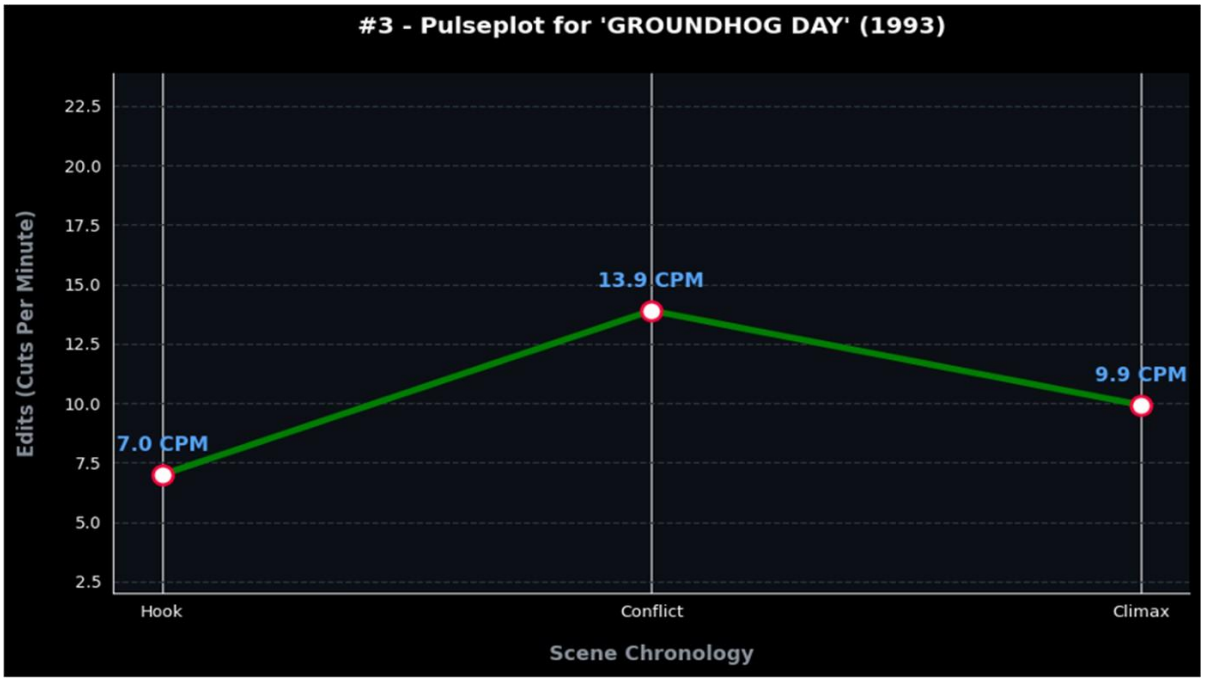

Figure 6.1.4: The Pulseplot (Sample shown for the movie - Groundhog Day (1993))

Interpretation:

In cinema, implementing edits (Cuts) is equivalent to providing energy to the particular sequence, to be more specific, the visual-oriented energy. As a part of analyzing the measure of edits using the metric ‘Cuts per Minute (CPM)’, for most of the movies, the editing technique seems to be implemented more in conflict and climax sequences, rather than the hook scenes. This denotes that the movie is at the peak in either of these sequences, or both, depending on the situation being depicted as well.

Objective 04: Era-specific Rhythm -

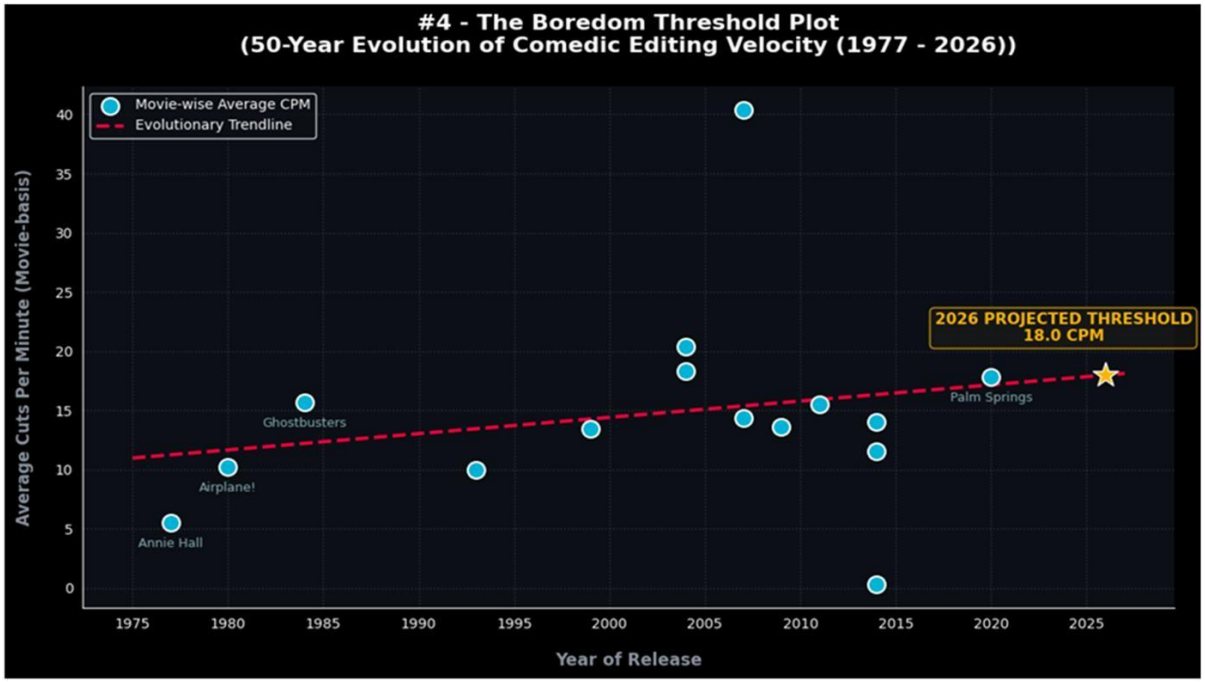

Figure 6.1.5: The Boredom Threshold Plot

Interpretation:

Though cinema is a resultant blend of multiple technical efforts, editing is one such crucial factor that gives shape to the mega output called movie, that is further experienced by the audience in the bigscreen or the miniscreen. Through an understanding of the era-specific editing-oriented pacing style for all the movies, it is evident that comedy is becoming visually faster and more "manic" over time. While older classics like Annie Hall (1977) and Airplane! (1980) had a relatively relaxed pace of 5 to 10 cuts per minute (CPM), modern comedies are trending toward a projected threshold of 18.0 CPM by 2026. This evolution suggests that filmmakers are increasingly relying on rapid editing to keep modern audiences engaged, moving away from the slower, dialogue-driven pacing of the past.

6.2. STATISTICAL MODELLING

Tool used: R Programming

Purpose:

To test the statistical significance of the difference between the technical variables and engineer the technical directive blueprint table, beneficial for the cinema technicians.

Coding workflow:

Loading the data set
(CSV format)

⇩

Data Pre-processing
(Tailoring the existing data through additional calculations and other techniques to proceed further)

⇩

Data splitting
(An era-wise analytical approach is being considered)

⇩

Determining the statistical significance
(Using Analysis of Variance (AOV) and Tukey test techniques for categorical-wise understanding (era-technology basis))

⇩

Probability prediction phase
(Building the model with all the variables (Predicted & Predictors) and determining the Odds Ratio for each category)

⇩

Framing a technical blueprint table
(Useful for the technicians to engineer cinematic tensions)

Figure 6.2.1: Statistical Modelling - workflow chart

Outputs:

Statistical significance -

```
# Test 01 - The Classic Cinema
# CPM & WPS
print("UNDERSTANDING THE CLASSIC CINEMA")

## [1] "UNDERSTANDING THE CLASSIC CINEMA"

pre_cpm_anova <- aov(cuts_per_minute ~ tension_cat, data = pre_2000)
summary(pre_cpm_anova)

##             Df Sum Sq Mean Sq F value Pr(>F)
## tension_cat  3  179.8   59.92   3.273 0.0627 .
## Residuals   11  201.4   18.31
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
TukeyHSD(pre_cpm_anova)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = cuts_per_minute ~ tension_cat, data = pre_2000)
##
## $tension_cat
##                       diff        lwr      upr     p adj
## Absurdity-Setup   4.042778  -6.471251 14.55681 0.6640490
## Cringe-Setup     11.790092  -1.086912 24.66710 0.0760466
## Payoff-Setup      7.906608  -2.867067 18.68028 0.1806133
## Cringe-Absurdity  7.747314  -2.766716 18.26134 0.1781887
## Payoff-Absurdity  3.863830  -3.933584 11.66124 0.4742692
## Payoff-Cringe    -3.883484 -14.657159  6.89019 0.7054376
```

```r
pre_wps_anova <- aov(words_per_second ~ tension_cat, data = pre_2000)
summary(pre_wps_anova)
```

```
##             Df Sum Sq Mean Sq F value Pr(>F)
## tension_cat  3  1.562  0.5207   0.991  0.433
## Residuals   11  5.782  0.5256
```

```r
TukeyHSD(pre_wps_anova)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = words_per_second ~ tension_cat, data = pre_2000)
##
## $tension_cat
##                         diff       lwr       upr     p adj
## Absurdity-Setup  -0.66504757 -2.446584 1.1164888 0.6834359
## Cringe-Setup     -0.95281908 -3.134747 1.2291084 0.5733005
## Payoff-Setup     -1.00269142 -2.828223 0.8228401 0.3910925
## Cringe-Absurdity -0.28777151 -2.069308 1.4937648 0.9605939
## Payoff-Absurdity -0.33764385 -1.658867 0.9835789 0.8666713
## Payoff-Cringe    -0.04987234 -1.875404 1.7756592 0.9997878
```

Figure 6.2.2: Statistical significance test - Pre-Millennium movies

```r
# Test 02 - The Modern Cinema
# CPM & WPS
print("UNDERSTANDING THE MODERN CINEMA")
```

```
## [1] "UNDERSTANDING THE MODERN CINEMA"
```

```r
post_cpm_anova <- aov(cuts_per_minute ~ tension_cat, data = post_2000)
summary(post_cpm_anova)
```

```
##             Df Sum Sq Mean Sq F value Pr(>F)
## tension_cat  3  162.3   54.09   0.494   0.69
## Residuals   26 2849.4  109.59
```

```
TukeyHSD(post_cpm_anova)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = cuts_per_minute ~ tension_cat, data = post_2000)
##
## $tension_cat
##                        diff        lwr      upr     p adj
## Absurdity-Setup   6.586340 -13.720806 26.89349 0.8101646
## Cringe-Setup      4.616704 -13.921099 23.15451 0.9025460
## Payoff-Setup      7.976237 -11.169523 27.12200 0.6669842
## Cringe-Absurdity -1.969636 -16.328957 12.38968 0.9814386
## Payoff-Absurdity  1.389897 -13.746156 16.52595 0.9942485
## Payoff-Cringe     3.359534  -9.304197 16.02326 0.8850804
```

```
post_wps_anova <- aov(words_per_second ~ tension_cat, data = post_2000)
summary(post_wps_anova)
```

```
##             Df Sum Sq Mean Sq F value Pr(>F)
## tension_cat  3  0.483  0.1609   0.554   0.65
## Residuals   26  7.546  0.2902
```

```
TukeyHSD(post_wps_anova)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = words_per_second ~ tension_cat, data = post_2000)
##
## $tension_cat
##                          diff        lwr       upr     p adj
## Absurdity-Setup   0.184518021 -0.8605465 1.2295825 0.9618901
## Cringe-Setup     -0.003098564 -0.9571076 0.9509105 0.9999997
## Payoff-Setup     -0.178893000 -1.1641893 0.8064033 0.9587851
## Cringe-Absurdity -0.187616584 -0.9265888 0.5513556 0.8974897
## Payoff-Absurdity -0.363411020 -1.1423561 0.4155341 0.5834453
## Payoff-Cringe    -0.175794436 -0.8275067 0.4759178 0.8800384
```

Figure 6.2.3: Statistical significance test - Post-Millennium movies

Interpretation:

From the statistical significance test conducted for all the cases featuring pre and post-millenium movies, a notable difference or a categorization across the tension categories cannot be observed, when Cuts Per Minute (Edits) or Words per Second (WPS) are considered. Since the p-values across all the cases are above the threshold limit (0.05), no statistical significance can be observed.

Probability prediction -

```r
# Model building - Edits (CPM) and Words (WPS) predicts the Tension category
prediction_model <- polr(tension_cat ~ cuts_per_minute + words_per_second,
                         data = hcd,
                         Hess = TRUE)
summary(prediction_model)

## Call:
## polr(formula = tension_cat ~ cuts_per_minute + words_per_second,
##     data = hcd, Hess = TRUE)
##
## Coefficients:
##                      Value Std. Error t value
## cuts_per_minute    0.04446    0.03256   1.365
## words_per_second -0.84310    0.49174  -1.715
##
## Intercepts:
##                   Value   Std. Error t value
## Setup|Absurdity  -2.8211  1.0655    -2.6478
## Absurdity|Cringe -1.0970  0.9727    -1.1278
## Cringe|Payoff     0.3193  0.9574     0.3335
##
## Residual Deviance: 113.0577
## AIC: 123.0577

# Odds-Ratio for easy interpretation
odds_ratio <- exp(coef(prediction_model))
print(odds_ratio)

##  cuts_per_minute words_per_second
##        1.0454629        0.4303757
```

Figure 6.2.4: Odds Ratio - Output

Interpretation:

The tension prediction model clearly shows that for every additional cut introduced, the progress in tension is more likely to happen by at least 4.5% (CPM = 1.045), while, for every additional word introduced, the progress in tension can be expected to decrease by 57% (0.430). This means that tension is highly progressive, when the visual engine (CPM) is more emphasized than the verbal engine (WPS).

Technical Directive table -

Technical Directive Table for Cinematic Tension Engineering

Era	Category	Target CPM	Target WPS	Technical Directive	Narrative Goal
Post-Millennium	Setup	10.8	1.43	Contextual Foundation	Prioritize verbal clarity; keep visual rhythm stable.
Post-Millennium	Absurdity	17.4	1.62	Rising Complexity	Increase technical energy; accelerate CPM by ~50%.
Post-Millennium	Cringe	15.4	1.43	Social Friction	Silence is key. Drop WPS to amplify audience discomfort.
Post-Millennium	Payoff	18.8	1.25	Rhythmic Climax	Peak Velocity. Maximize CPM; let visual cuts drive humor.
Pre-Millennium	Setup	4.9	2.19	Contextual Foundation	Prioritize verbal clarity; keep visual rhythm stable.
Pre-Millennium	Absurdity	9.0	1.53	Rising Complexity	Increase technical energy; accelerate CPM by ~50%.
Pre-Millennium	Cringe	16.7	1.24	Social Friction	Silence is key. Drop WPS to amplify audience discomfort.
Pre-Millennium	Payoff	12.8	1.19	Rhythmic Climax	Peak Velocity. Maximize CPM; let visual cuts drive humor.

Figure 6.2.5: Technical Directive table - Output

Interpretation:

The technical directive blueprint table shows a clear distinction between the quantifiable techniques to be implemented. As the era changes, the target CPM is seen to be highly demanded, as the movie progresses across the tension level of 'Setup' (majorly the introduction sequence) to the 'Big payoff' (majorly the climax sequence). Such a distinction is clearly seen in the target CPM of 4.9 corresponding to the Setup in pre-millennium movies, while the same is at 10.8 CPM target level in post-millennium movies. In the case of WPS, it is shown in a less intensifying manner. This clearly highlights the current dominance of rapid cuts over the long dialogues to show effectiveness in conveying humor amongst the cinema lovers.

6.3. MACHINE LEARNING

Tool used: Python (Google Collab)

Link: [MDA_Project_ML.ipynb]

Purpose:

To predict the Tension score based on the CPM & WPS provided. This helps the technicians to decide on the pacing setup (words or edits) that can lead the audience to experience a particular category of tension across a comedy movie.

Coding workflow:

Importing the necessary libraries
(NumPy, Pandas, SkLearn)

⇩

Loading the data set
(CSV format)

⇩

Adding new metrics
(Cuts per Minute (CPM) & Words per Second (WPS))

⇩

Dividing the data
(Input - CPM & WPS / Output - Tension score)

⇩

Train-test splitting phase
(80/20 proportion)

⇩

Model training phase
(Technique - Random Forest Regressor)

⇩

Model prediction and evaluation
(Evaluators - Mean Absolute Error (MAE) & Mean Squared Error (MSE)

⇩

Prediction architecture creation
(Building a function to conduct the prediction)

⇩

Generating the prediction output
(Calling the function)

Figure 6.3.1: Prediction using Machine Learning - workflow chart

Output:

```
# Prediction using the function
tension_prediction(45.0,4.0)
```

```
--- Prediction Results ---
Input Pacing Setup -> CPM: 45.0, WPS: 4.0
Predicted Raw Tension Score: 4.68
Tension Category: Rising Absurdity - Some sort of misunderstanding is growing
```

Figure 6.3.2: Predicting the tension category - Output

Interpretation:

The Machine Learning prediction phase of this project utilizes a Random Forest Regressor (RFR) to bridge the gap between technical film metrics and narrative impact by predicting audience Tension Scores based on Cuts Per Minute (CPM) and Words Per Second (WPS). While the model currently operates with a Mean Absolute Error (MAE) score of 2.09, it successfully translates raw technical data into meaningful cinematic stages. A high-velocity pacing setup of 45 CPM and 4 WPS is predicted as a "Rising Absurdity" score of 4.68, signaling a growing narrative misunderstanding rather than a final climax. For editors and directors, this tool provides an action-oriented framework for "tension engineering," allowing them to precisely calibrate the balance between visual cuts and dialogue density to ensure a scene hits its intended emotional category, such as increasing CPM to drive a "Big Payoff" or reducing WPS to amplify "Cringe" through uncomfortable silence.

6.4. DASHBOARD

Tool used: Tableau Public

Link: [Comedy_Cadence_Project_BT | Tableau Public]

Purpose:

To present the patterns and insights in such a way that, it would create an effective understanding amongst the non-technical batch of users and stakeholders.

Dashboard:

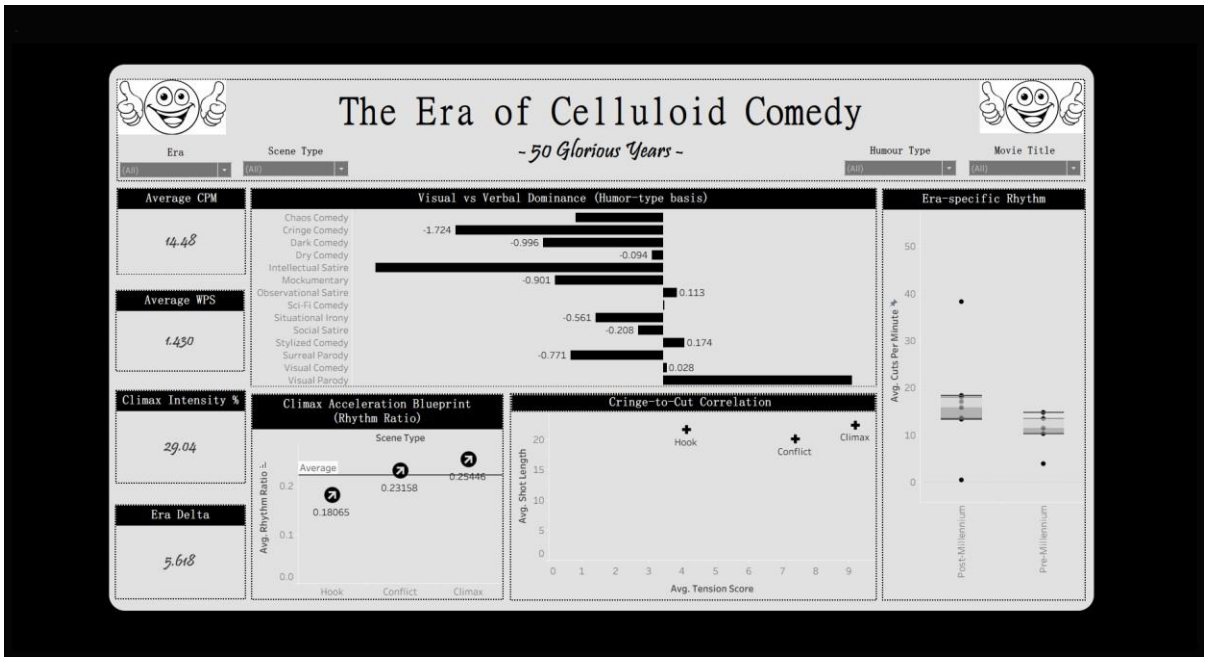

Figure 6.4.1: The Dashboard

Interpretation -

Key Performance Indicators (KPI):

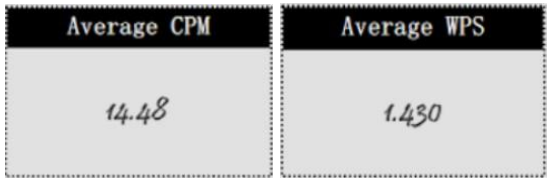

Figure 6.4.2: Global KPIs - Average Cuts per Minute & Average Words per Second

Average CPM and Average WPS are the global KPIs. The current benchmark denotes an average of nearly 15 Cuts per Minute in case of Editing, and nearly 2 Words per Second in case of Dialogue writing. If the so-called metric (either CPM or WPS) shows an increase (When the live metric measure is compared with the prevailing benchmark), while one increases and the other remains constant, a case of over-writing (when WPS > CPM) or over-editing (when CPM > WPS) can be considered. It is important to balance both the aspects accordingly. Hence, this insight majorly benefits the Screen-Writers and the Sound Editors to find a sync with each other.

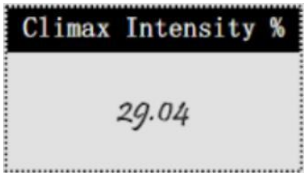

Figure 6.4.3: Editor special KPI - Climax Intensity (%)

Climax intensity shift (%) is specially dedicated to the progressive editing mechanism under the light of a movie. A shortfall in the current metric measurement with respect to the benchmark, may cause the scene to be less energetic. A minimum of 29% increase in Cuts Per Minute (CPM) must be considered to make an impactful cinematic progress nowadays. Hence, this insight majorly benefits the Directors and the Creative Editors to know the ideal pacing mechanism of a movie.

Figure 6.4.4: Era-specific KPI - Era Delta

Era pacing delta specializes in comparing the average editing oriented mechanism, differentiated on the basis of era (Pre-Millennium/Post-Millennium). Through the benchmark of 5.618, it denotes that, the modern films are edited by at least 5.6 CPM rate in excess. If that old format movie experience is still needed again, this benchmark can be deducted. Hence, this insight majorly helps the Creative Director and the Colorist, in order to design a sequence in such a way that it would cope up with the era-specific style of story telling.

Charts:

Visual vs. Verbal Dominance -

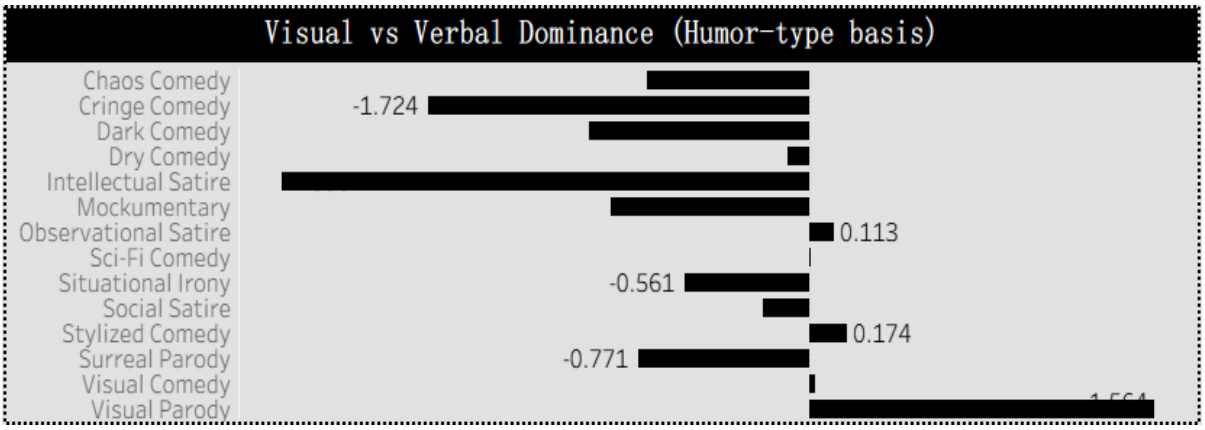

Figure 6.4.5: Barplot - Visual vs. Verbal Dominance

Provided the mode of conveying the humor is purely a subjective matter under the light of a specific humor type, as seen across most of the humor types, though intensity is quite lesser, words are preferred more than the visuals in conveying the humor (Verbal dominance is frequent). But era-wise, the distinction in the level of intensity of such a dominance can be noted. This helps the technicians (Director/Editor) to create a humor-specific film, with a clear idea on the dominance factor to be considered.

Cringe-to-Cut Correlation -

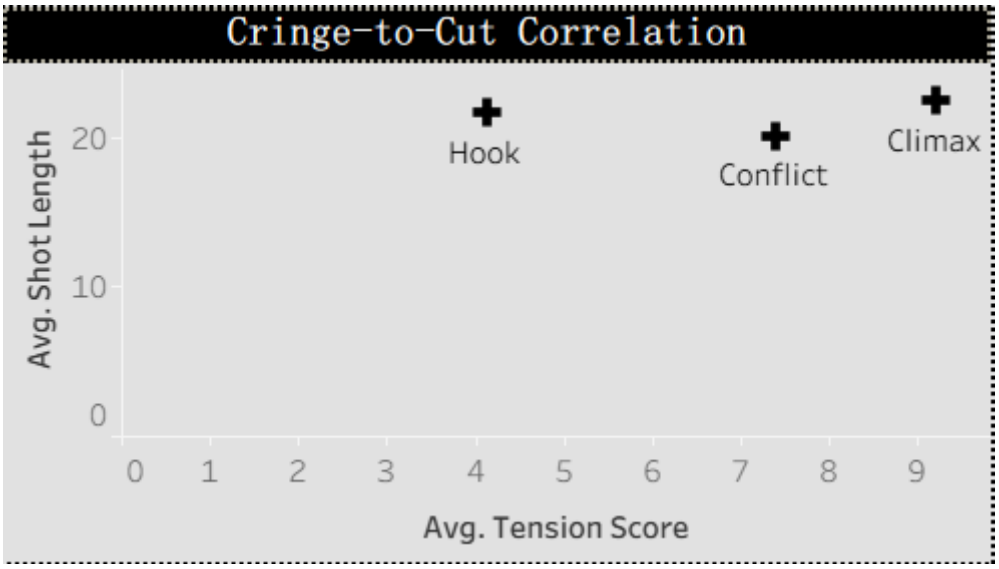

Figure 6.4.6: Scatterplot - Cringe-to-Cut Correlation

In cinema, 'Cringe' is a state where the audience is forced to feel the tension of a sequence, without any interruption being noted visually. Here it means that, as and when the camera

shots are prolonged, the number of cuts or edits are being avoided to create tension amongst the audience. Throughout the pre-millennium movies, the cringe factor, though not on a heavy note, can be mostly felt in the conflict scenes (Absurdity (Tension score of 6) is created when the camera is lingered for nearly 12 seconds, on an average). Throughout the post-millennium movies, the cringe factor can be felt on a heavy note in the climax scenes (Big payoff (Tension score of 9) is created when the camera is lingered for nearly 31 seconds, on an average). The intensity of words or silence can be felt throughout such sequences, when editing is not an existing factor. This can help the directors and editors to determine, where to implement the editing mechanism, so that a highly energetic tension can be felt by the audience through a scene or a sequence.

Climax Acceleration Blueprint -

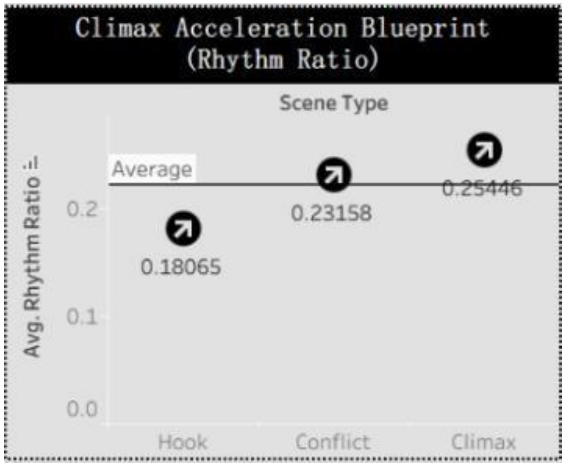

Figure 6.4.7: Graph - Climax Acceleration Blueprint

The climax acceleration blueprint majorly focuses on the Rhythm Ratio, that provides an insight on the intensity of Visual engine (Cuts) over the Verbal engine (Words). With respect to the position of global average of Rhythm Ratio at 0.22, from hook scene, where the verbal engine is active (Rhythm Ratio < Global Average), to the conflict, then to the climax scene, the Visual engine seems to be at the track. This clearly shows that the humor and its resultant effects are felt to be highly impactful, when visuals speak more than the words. This helps the modern filmmakers to get a clarity on the standard scale of editing to be implemented across the scenes, as the movie sees the progress.

Era-specific Rhythm -

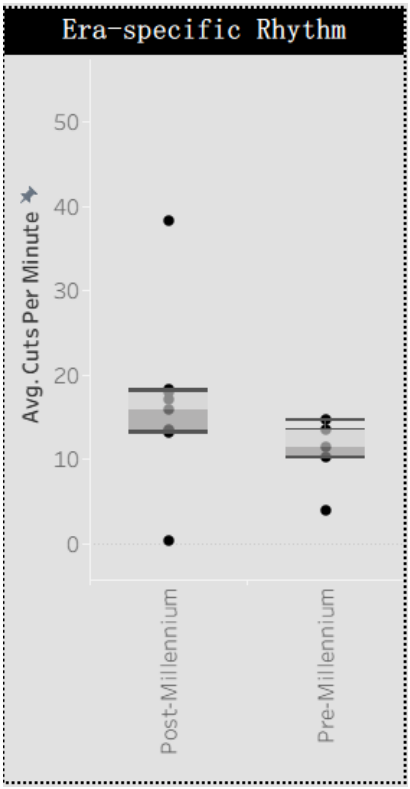

Figure 6.4.8: Boxplot - Era-specific Rhythm

On an era-specific note, the boxplot shows a clear picture of Cuts Per Minute (CPM) at a higher position in the post-millennium movies, having ranged between 13 CPM to 18 CPM, comparing to the pre-millennium movies, showing a range of 10 CPM to 13 CPM. This insight helps an editor to understand the style and standard of placing cuts across the scenes, that demands an era-specific style of storytelling. This means that, if a director is interested in attempting the classical style of movie making, an editor can limit the number of cuts accordingly, and vice versa.

7. CONCLUSION

Cinema has found its irreplaceable seat in the world of entertainment. For the world is so busy that the stressbusters are constantly needed to be in place, cinema and its humor will always find its prominence.

The past cinematic technology had its own limitations and constraints, especially in case of editing. While chronological-based footage editing was followed by the olden-era editors, consuming a huge amount of time and effort, the current advancement of technology has favoured the evolution of modern cinematic technology as well, especially in the form of Non-Linear Editing (NLE) technology, that allows for a massive editing (for instance, up to

50 cuts per second rate) from anywhere, anytime. The evolution of Artificial Intelligence is another remarkable automation solution in making things much easier.

As the scenario moves on, it is very crucial to understand that, though editing was implemented to pace the humor towards the climax in the movies before the 2000s, the intensity of editing has increased in the current cinematic era, especially in this context. Not only in terms of editing, but also in terms of dialogues, the impact of humor can only be brought in well, when it is built meaningfully powerful. The performance driven humor is truly appreciated, but the current pattern of audiences' tastes prefer for a more visually energetic

In such an aspect, this project features a data-oriented solution to the real and existing technical-wise aspects and the related confusion that are confronted by the technical team, especially in the most crucial phase of the post-production. Hence, the outputs from this project can aid the major cinematic technicians like directors, script writers, editors, and cinematographers in providing a clarity on framing the sequences that are ideally balanced in terms of technology (Cuts, Shot lengths, and Dialogues), aiming at conveying the humor at its maximum impact.

8. LIMITATION AND FURTHER RECOMMENDATION

There are numerous genres that continue to create high impact in the film industry. But this project features only the 'Comedy' genre. As reflected through the results from the statistical modelling phase, especially in the Analysis of Variance (AOV) and Post Tukey HSD tests, where the statistical significance could not be found (p-value > 0.05), the limitation also pertains to the sample size considered for this project (15 movies). From the results from the machine learning prediction phase, it is evident that the model serves as a prototype or as a baseline, since only the pacing of comedy-oriented tension is being studied here (MAE - 2.09 & MSE - 6.42). While edits and words correlate with the tension categories, the intensity of tension can further be investigated using other factors like sentimental analysis (tone of dialogues), music (BPM), and so on.

While this project can serve as an effective benchmark, considering all the factors under limitation, a scope for further studies, especially related to other genres, can be considered, as it can be taken to a larger scale, with more samples and the relevant features at stake.

9. CODE ATTACHMENTS

Exploratory Data Analysis (EDA):

Importing the necessary libraries -

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Loading the data set -

```
hollywood_cinema_data = pd.read_csv('/content/Project_Master_Data.csv')
```

Conducting a preliminary check-up of the data -

```
hollywood_cinema_data.head(6)
```

	Movie_Title	Released_Year	Humour_Type	Scene_Type	Duration_Sec	Total_Cuts	Word_Count	Tension_Score	Notes
0	Annie Hall	1977	Intellectual Satire	Hook	105	1	304	1	A simple verbal introduction of the main chara...
1	Annie Hall	1977	Intellectual Satire	Conflict	80	2	227	6	Tension is arisen within a funny ambience, whe...

```
hollywood_cinema_data.shape
```

(45, 9)

```
hollywood_cinema_data.index
```

RangeIndex(start=0, stop=45, step=1)

```
hollywood_cinema_data.columns
```

Index(['Movie_Title ', 'Released_Year', 'Humour_Type', 'Scene_Type', 'Duration_Sec', 'Total_Cuts', 'Word_Count', 'Tension_Score', 'Notes'], dtype='object')

```
hollywood_cinema_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 45 entries, 0 to 44
Data columns (total 9 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Movie_Title    45 non-null     object
 1   Released_Year  45 non-null     int64
 2   Humour_Type    45 non-null     object
 3   Scene_Type     45 non-null     object
 4   Duration_Sec   45 non-null     int64
 5   Total_Cuts     45 non-null     int64
 6   Word_Count     45 non-null     int64
 7   Tension_Score  45 non-null     int64
 8   Notes          45 non-null     object
dtypes: int64(5), object(4)
memory usage: 3.3+ KB
```

```
hollywood_cinema_data.describe()
```

	Released_Year	Duration_Sec	Total_Cuts	Word_Count	Tension_Score
count	45.000000	45.000000	45.000000	45.000000	45.000000
mean	2002.466667	294.155556	78.244444	425.733333	6.911111
std	12.960780	195.867529	106.937993	327.126011	2.521023
min	1977.000000	44.000000	1.000000	47.000000	1.000000
25%	1993.000000	188.000000	39.000000	227.000000	5.000000
50%	2007.000000	265.000000	66.000000	352.000000	8.000000
75%	2014.000000	346.000000	85.000000	494.000000	9.000000
max	2020.000000	1055.000000	728.000000	1511.000000	10.000000

```
hollywood_cinema_data.describe(include="object")
```

	Movie_Title	Humour_Type	Scene_Type	Notes
count	45	45	45	45
unique	15	14	3	45
top	Annie Hall	Cringe Comedy	Hook	A simple verbal introduction of the main chara...
freq	3	6	15	1

Objective 01: Visual vs. Verbal Dominance -

Creating the function and calculating the metrics:

```
def objective1(data,movie_title):
```

```
  data.columns = data.columns.str.strip() # To remove the trail spaces
in column heads
  movie_data = data[data['Movie_Title'].str.strip() == movie_title] #
Filtering for the movie name in the data set

  if movie_data.empty:
    print('No Movies found. Please check the spelling.') # If in case
there is a mismatch in the movie title
    return

  released_year = movie_data['Released_Year'].iloc[0] # Retrieving the
year of release for a movie

  print(f"Movie - {movie_title.upper()} ({released_year})") # Main
header section
  print('*'*65) # Seperation line, for easy readability

  for index, rows in movie_data.iterrows(): # Iteration through the
needed features
    scene = rows['Scene_Type']
    duration = rows['Duration_Sec']
    cuts = rows['Total_Cuts']
    words = rows['Word_Count']

    # Metrics 01 - Rhythm Ratio (rr)
    rr = cuts / words if words > 0 else 0 # Just a Provision, if in
case the value goes to 0

    # Metrics 02 - Words Per Second (wps)
    wps = words / duration if duration > 0 else 0 # Just a Provision,
if in case the value goes to 0

    # Metrics 03 - Cuts Per Minute (cpm)
    cpm = (cuts / duration) * 60 if duration > 0 else 0 # Duration (in
Seconds), now converted to Minutes (*60)

    if rr < 0.10 and wps > 1.5: # Benchmarking the metrics for the
clear-cut insights
      dominance_style = 'VERBAL DOMINANCE'
      observation = 'Sorkin-style approach - More WORDS, less CUTS'
```

```
    elif 0.10 <= rr <= 0.25 and cpm > 10:
      dominance_style = 'BALANCED RHYTHM'
      observation = 'A great synergy between the WORDS and the CUTS'
    elif rr > 0.25 and cpm > 12:
      dominance_style = 'VISUAL DOMINANCE'
      observation = 'Wright-style approach - More CUTS, less WORDS'
    else:
      dominance_style = 'STAGNANT'
      observation = 'A slower-pace between the WORDS and the CUTS'

    print(f"SCENE: {scene.upper()}") # For an organised way of
displaying the output
    print(f"Rhythm Ratio: {rr:.3f} (Cuts per Word)")
    print(f"Delivery Pace: {wps:.2f} W/s (Words per Second)")
    print(f"Edit Velocity: {cpm:.2f} CPM (Cuts per Minute)")
    print(f"VERDICT: {dominance_style}")
    print(f"INSIGHT: {observation}")
    print("-"*65) # A seperation line for easy readability

objective1(hollywood_cinema_data,'Airplane!')
```

```
Movie - AIRPLANE! (1980)
*****************************************************************
SCENE: HOOK
Rhythm Ratio: 0.104 (Cuts per Word)
Delivery Pace: 1.49 W/s (Words per Second)
Edit Velocity: 9.26 CPM (Cuts per Minute)
VERDICT: STAGNANT
INSIGHT: A slower-pace between the WORDS and the CUTS
-----------------------------------------------------------------
SCENE: CONFLICT
Rhythm Ratio: 0.076 (Cuts per Word)
Delivery Pace: 1.84 W/s (Words per Second)
Edit Velocity: 8.40 CPM (Cuts per Minute)
VERDICT: VERBAL DOMINANCE
INSIGHT: Sorkin-style approach - More WORDS, less CUTS
-----------------------------------------------------------------
SCENE: CLIMAX
Rhythm Ratio: 0.213 (Cuts per Word)
Delivery Pace: 1.28 W/s (Words per Second)
Edit Velocity: 16.42 CPM (Cuts per Minute)
VERDICT: BALANCED RHYTHM
INSIGHT: A great synergy between the WORDS and the CUTS
-----------------------------------------------------------------
```

Plotting the chart:

```
def cinema_trajectory_map(data, movie_title):
```

```
  movie_name = movie_title.strip()

  target_movie = data[data['Movie_Title'].str.strip() ==
movie_name].copy() # Considering only the movie under the limelight
  released_year = target_movie['Released_Year'].iloc[0]
  others = data[data['Movie_Title'].str.strip() != movie_name] #
Considering other movies

  scene_order = {'Hook': 0, 'Conflict': 1, 'Climax': 2} # Target movie
segregated by scene-order
  target_movie['order'] = target_movie['Scene_Type'].map(scene_order)
  target_movie = target_movie.sort_values('order')

  # Creating the plot
  plt.figure(figsize=(10,5))
  sns.set_style("darkgrid")


  plt.scatter(others['Word_Count'], others['Total_Cuts'], #
Specifications for other movies
              alpha=0.2, c='gray', label='Industry Baseline')

  plt.plot(target_movie['Word_Count'], target_movie['Total_Cuts'], #
Specifications for the target movie
           marker='o', markersize=10, linewidth=3, color='Red',
           label=f'Path of {movie_name}')

  for i, row in target_movie.iterrows(): # Scene-wise information
marking
        plt.annotate(f"{row['Scene_Type']}", (row['Word_Count'],
row['Total_Cuts']),
                     textcoords="offset points", xytext=(5,5),
fontweight='bold', fontsize=10, color='#1d3557') # Setting up the
aesthetics

  plt.title(f"#1 - Visual-Verbal Trajectory of {movie_name}
({released_year})", color='black', fontsize=15, fontweight='bold',
pad=20)
  plt.xlabel("Verbal Engine (Word Count)", fontsize=12)
  plt.ylabel("Visual Engine (Total Cuts)", fontsize=12)
```

```
  plt.legend()

  plt.text(data['Word_Count'].max()*0.7, data['Total_Cuts'].min(),
"Verbal Zone", alpha=0.5, fontsize=12, color='green') # Marking the
dominance zone
  plt.text(data['Word_Count'].min(), data['Total_Cuts'].max()*0.9,
"Visual Zone", alpha=0.5, fontsize=12, color='red')
  plt.tight_layout()

  plt.show()

cinema_trajectory_map(hollywood_cinema_data,'Airplane!')
```

Output: *Figure 6.1.2: The Trajectory Map*

Objective 02: Cringe-to-Cut correlation -

Creating the function and calculating the metrics:

```
def objective2(data, movie_title):
  movie_name = movie_title.strip()
  movie_data = data[data['Movie_Title'].str.strip() == movie_name]

if movie_data.empty:
    print(f"No Movies found. Please check the spelling.")
    return

released_year = movie_data['Released_Year'].iloc[0]

  print(f"Movie - {movie_name.upper()} ({released_year})")
  print('*'*85)

for index, row in movie_data.iterrows():
    scene = row['Scene_Type']
    duration = row['Duration_Sec']
    cuts = row['Total_Cuts']
    tension = row['Tension_Score']

# Metric 01 - Average Shot Length (asl)
    asl = duration / cuts if cuts > 0 else duration
```

```
# Metric 02 - Cuts Per Minute (cpm)
    cpm = (cuts / duration) * 60 if duration > 0 else 0

# Benchmarking the metrics
    if asl >= 6.0 and tension >= 6:
      verdict = "CRINGE TENSION"
      observation = "The director holds the shot intentionally long to
make the audience feel the awkwardness or dread."
    elif asl < 4.0 and tension >= 6:
      verdict = "PANIC TENSION"
      Observation  = "High tension is conveyed through rapid, chaotic
editing rather than lingering discomfort."
    elif asl >= 6.0 and tension < 6:
      verdict = "ATMOSPHERIC SETUP"
      observation = "Low tension with long shots. The scene is likely
building world-context or casual character development."
    else:
      verdict = "STANDARD PACING"
      observation = "Neutral comedic pacing. Neither heavily rushed nor
intentionally awkward."

    print(f"SCENE: {scene.upper()}")
    print(f"Tension Score: {tension}/10")
    print(f"Average Shot Length: {asl:.2f} Seconds")
    print(f"Edit Velocity: {cpm:.2f} Cuts per Minute")
    print(f"VERDICT: {verdict}")
    print(f"INSIGHT: {observation}")
    print("-"*85)

objective2(hollywood_cinema_data,'Mean Girls')
```

```
•••  Movie - MEAN GIRLS (2004)
     ********************************************************************************
     SCENE: HOOK
     Tension Score: 4/10
     Average Shot Length: 2.65 Seconds
     Edit Velocity: 22.60 Cuts per Minute
     VERDICT: STANDARD PACING
     INSIGHT: Neutral comedic pacing. Neither heavily rushed nor intentionally awkward.
     --------------------------------------------------------------------------------
     SCENE: CONFLICT
     Tension Score: 8/10
     Average Shot Length: 4.34 Seconds
     Edit Velocity: 13.82 Cuts per Minute
     VERDICT: STANDARD PACING
     INSIGHT: Neutral comedic pacing. Neither heavily rushed nor intentionally awkward.
     --------------------------------------------------------------------------------
     SCENE: CLIMAX
     Tension Score: 9/10
     Average Shot Length: 3.43 Seconds
     Edit Velocity: 17.48 Cuts per Minute
     VERDICT: PANIC TENSION
     INSIGHT: Neutral comedic pacing. Neither heavily rushed nor intentionally awkward.
     --------------------------------------------------------------------------------
```

Plotting the chart:

```
def cringe_heatmap(data):

  data['asl'] = data['Duration_Sec'] / data['Total_Cuts']

  plt.style.use('dark_background')
  fig, ax = plt.subplots(figsize=(10,10))
  ax.set_facecolor('#0a0a0a') # This code denotes a deep theatrical
black color

# Setting up the heatmap architecture
  sns.kdeplot(
        x=data['Tension_Score'],
        y=data['asl'],
        cmap="inferno",
        fill=True,
        thresh=0.05,
        bw_adjust=0.85,
        ax=ax,
        alpha=0.85)

# Plotting the scatterplot
```

```
  sns.scatterplot(
        x=data['Tension_Score'],
        y=data['asl'],
        color='white',
        s=40,
        edgecolor='#aaaaaa',
        alpha=0.9,
        ax=ax,
        zorder=5)

tension_threshold = 6.0 # Threshold lines
  asl_threshold = 5.0
  plt.axvline(x=tension_threshold, color='white', linestyle='--',
alpha=0.3, zorder=4)
  plt.axhline(y=asl_threshold, color='white', linestyle='--',
alpha=0.3, zorder=4)

# Zone labelling for easy identification
  plt.text(tension_threshold + 0.3, asl_threshold + 1.5, "THE CRINGE
ZONE (High Tension, Lingering Camera)",
             color='Black', fontsize=8, fontweight='bold', zorder=6)

  plt.text(tension_threshold + 0.3, asl_threshold - 1.5, "PANIC (High
Tension, Rapid Cuts)",
             color='Blue', fontsize=8, fontweight='bold', zorder=6)

  plt.text(1, asl_threshold + 1.5, "DEADPAN (Low Tension, Slow Cuts)",
             color='White', fontsize=8, fontweight='bold', alpha=0.7,
zorder=6)

  plt.title("#2 - The Cringe Heatmap\n(Correlation between Narrative
Tension and the Linger Factor)",
              fontsize=16, fontweight='bold', color='white', pad=20)
  plt.xlabel("Reported Stakes (Tension Score: 1-10)", fontsize=12,
labelpad=10, color='#f1faee')
  plt.ylabel("The Linger Factor (Average Shot Length in Sec)",
fontsize=12, labelpad=10, color='#f1faee')

# Boundary setting for a balanced graph display
  plt.xlim(0, 11)
```

```
  plt.ylim(0, data['asl'].max() + 2)

  plt.grid(False)
  plt.tight_layout()

  plt.show()

cringe_heatmap(hollywood_cinema_data)
```

Output: *Figure 6.1.3: The Cringe Heatmap*

Objective 03: The Climax Acceleration Blueprint -

Creating the function and plotting the chart:

```
def pulseplot(df, movie_title):

    movie_name = movie_title.strip() # Information considered from the
main data (hollywood_cinema_data)
    movie_data = df[df['Movie_Title'].str.strip() == movie_name].copy()
    released_year = movie_data['Released_Year'].iloc[0]

    # Metric considered - Cuts per Minute (cpm)
    movie_data['cpm'] = (movie_data['Total_Cuts'] /
movie_data['Duration_Sec']) * 60

    # Mapping the scene order as per the chronology (Hook - Conflict -
Climax)
    scene_order = {'Hook': 1, 'Conflict': 2, 'Climax': 3}
    movie_data['story_order'] =
movie_data['Scene_Type'].str.strip().map(scene_order)
    movie_data = movie_data.sort_values('story_order')

    plt.style.use('dark_background') # Plotting phase - Background
setup
    fig, ax = plt.subplots(figsize=(10, 6))
    ax.set_facecolor('#0d1117') # Deep studio color

    # Pulseplot - Aesthetics
    ax.plot(
```

```
        movie_data['Scene_Type'],
        movie_data['cpm'],
        color='Green',
        marker='o',
        markersize=12,
        linewidth=4,
        markerfacecolor='white',
        markeredgecolor='#ff003c',
        markeredgewidth=2,
        zorder=3)

    for index, row in movie_data.iterrows():
      ax.annotate(
          f"{row['cpm']:.1f} CPM",
            (row['Scene_Type'], row['cpm']),
            textcoords="offset points",
            xytext=(0, 15),
            ha='center',
            fontsize=12,
            fontweight='bold',
            color='#58a6ff')

    plt.title(f"#3 - Pulseplot for '{movie_name.upper()}'
({released_year})",
              fontsize=14, fontweight='bold', color='white', pad=25)
    plt.xlabel("Scene Chronology", fontsize=12, labelpad=15,
color='#8b949e', fontweight='bold')
    plt.ylabel("Edits (Cuts Per Minute)", fontsize=12, labelpad=15,
color='#8b949e', fontweight='bold')

    y_min, y_max = movie_data['cpm'].min(), movie_data['cpm'].max() #
Y-Axis threshold
    plt.ylim(max(0, y_min - 5), y_max + 10)

    ax.grid(axis='y', color='#30363d', linestyle='--', linewidth=1,
zorder=1) # Seperation grid
    ax.spines['top'].set_visible(False)
    ax.spines['right'].set_visible(False)

    plt.tight_layout()
```

```
    plt.show()

pulseplot(hollywood_cinema_data,'Groundhog Day')
```

Output: *Figure 6.1.4: The Pulseplot*

Objective 04: Era-specific Rhythm -

Creating the function and plotting the chart:

```
def era_boredom_threshold(df):

  # Aggregating Edits and Duration for each movies
  movie_era_data = df.groupby(['Movie_Title',
'Released_Year']).agg({'Total_Cuts': 'sum','Duration_Sec':
'sum'}).reset_index()

  # Metric considered - Movie-wise Cuts Per Minute (mcpm)
  movie_era_data['mcpm'] = (movie_era_data['Total_Cuts'] /
movie_era_data['Duration_Sec']) * 60
  movie_era_data = movie_era_data.sort_values('Released_Year')

  plt.style.use('dark_background') # Setting up the background
  fig, ax = plt.subplots(figsize=(12, 7))
  ax.set_facecolor('#0d1117')

  # Considering the year-wise historical CPM for plotting
  sns.scatterplot(
      data=movie_era_data,
      x='Released_Year',
      y='mcpm',
      s=150,
      color='#00b4d8',
      edgecolor='white',
      linewidth=1.5,
      ax=ax,
      zorder=3,
      label='Movie-wise Average CPM')

  x = movie_era_data['Released_Year'] # Statistical trendline
```

```
  y = movie_era_data['mcpm']

  z = np.polyfit(x, y, 1) # Line of best fit
  p = np.poly1d(z)

  future_years = np.arange(x.min() - 2, 2028) # Timeline extension
(Until the year 2026)

  # Plotting the trendline
  ax.plot(
      future_years,
      p(future_years),
      color='#ff003c',
      linestyle='--',
      linewidth=2.5,
      zorder=2,
      label='Evolutionary Trendline')

  # Decorating the 2026 prediction part
  forecast_2026 = p(2026)
  ax.scatter(2026, forecast_2026, color='#ffb703', s=350, marker='*',
edgecolor='white', zorder=4)

  ax.annotate(
      f"2026 PROJECTED THRESHOLD\n{forecast_2026:.1f} CPM",
      (2026, forecast_2026),
      textcoords="offset points",
      xytext=(-30, 25),
      ha='center',
      fontsize=11,
      fontweight='bold',
      color='#ffb703',
      bbox=dict(boxstyle="round,pad=0.3", fc="#1a1a1a", ec="#ffb703",
alpha=0.8))

  # Annotating Old vs New movies for easy identification
  for index, row in movie_era_data.iterrows():
    if row['Released_Year'] < 1985 or row['Released_Year'] > 2018:
      ax.annotate(
          row['Movie_Title'].strip(),
```

```
                (row['Released_Year'], row['mcpm']),
                textcoords="offset points",
                xytext=(0, -18),
                ha='center',
                fontsize=9,
                color='#a8dadc',
                alpha=0.8)

  # Main plot
  plt.title("#4 - The Boredom Threshold Plot\n(50-Year Evolution of
Comedic Editing Velocity (1977 - 2026))",
                fontsize=16, fontweight='bold', color='white', pad=25)
  plt.xlabel("Year of Release", fontsize=12, labelpad=15,
color='#8b949e', fontweight='bold')
  plt.ylabel("Average Cuts Per Minute (Movie-basis)", fontsize=12,
labelpad=15, color='#8b949e', fontweight='bold')

  ax.grid(axis='both', color='#30363d', linestyle=':', linewidth=1,
zorder=1) # Additional features
  ax.spines['top'].set_visible(False)
  ax.spines['right'].set_visible(False)
  plt.xticks(np.arange(1975, 2030, 5))
  ax.legend(loc='upper left', facecolor='#0d1117', edgecolor='white',
fontsize=10)

  plt.tight_layout()
  plt.show()

era_boredom_threshold(hollywood_cinema_data)
```

Output: *The Boredom Threshold Plot*

STATISTICAL MODELLING -

Loading the data set -

```
hcd <- read.table(file="clipboard",sep="\t",header=T)
head(hcd)
```

```
##   Movie_Title Released_Year         Humour_Type Scene_Type Duration_Sec
## 1  Annie Hall          1977 Intellectual Satire       Hook          105
## 2  Annie Hall          1977 Intellectual Satire   Conflict           80
## 3  Annie Hall          1977 Intellectual Satire     Climax          208
## 4   Airplane!          1980      Surreal Parody       Hook          188
## 5   Airplane!          1980      Surreal Parody   Conflict          821
## 6   Airplane!          1980      Surreal Parody     Climax          274
##   Total_Cuts Word_Count Tension_Score
## 1          1        304             1
## 2          2        227             6
## 3         33        414             9
## 4         29        280             3
## 5        115       1511             6
## 6         75        352             9
##
Notes
## 1                                                    A simple verbal introd
uction of the main character.
## 2                   Tension is arisen within a funny ambience, where the s
ociety is not being involved.
## 3                      Verbally presented with lot of cuts and dialogues,
but with intense word usages.
## 4                                     Characters are introduced with less
tension, in a more funny way.
## 5 Long sequence - Tension is high, but balanced with literal humour, invol
```

```
ving more cuts and dialogues.
## 6                          Breath-taking sequence, involving more of the vi
sual cuts than the dialogues.
```

Data Pre-processing -

library(dplyr)

##
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':
##
filter, lag

The following objects are masked from 'package:base':
##
intersect, setdiff, setequal, union

```
colnames(hcd) <- trimws(colnames(hcd)) # Removing spaces, if any
 hcd <- hcd %>%
   mutate(
          # Metrics
          cuts_per_minute = (Total_Cuts / Duration_Sec) *60,
          words_per_second = Word_Count / Duration_Sec,
          # Defining the era-wise splitting
          era = ifelse(Released_Year < 2000, "Pre-Millennium", "Post-Millennium"),
          # Categorizing the tension scores
          tension_cat = cut(Tension_Score,
```

```
                breaks <- c(0,3,6,8,10),
                labels <- c("Setup", "Absurdity", "Cringe", "Payoff"),
                include.lowest = TRUE)
  )
```

Data splitting -

```
pre_2000 <- filter(hcd,era=="Pre-Millennium")
post_2000 <- filter(hcd,era=="Post-Millennium")
```

Determining the statistical significance -

```
# Test 01 - The Classic Cinema
 # CPM & WPS
 print("UNDERSTANDING THE CLASSIC CINEMA")
```

```
## [1] "UNDERSTANDING THE CLASSIC CINEMA"
```

```
pre_cpm_anova <- aov(cuts_per_minute ~ tension_cat, data = pre_2000)
 summary(pre_cpm_anova)
```

```
##            Df Sum Sq Mean Sq F value Pr(>F)
 ## tension_cat  3  179.8   59.92   3.273 0.0627 .
 ## Residuals  11  201.4   18.31
 ## ---
 ## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
TukeyHSD(pre_cpm_anova)
```

```
##  Tukey multiple comparisons of means
 ##      95% family-wise confidence level
 ##
 ## Fit: aov(formula = cuts_per_minute ~ tension_cat, data = pre_2000)
 ##
 ## $tension_cat
 ##                          diff      lwr      upr     p adj
 ## Absurdity-Setup   4.042778  -6.471251 14.55681 0.6640490
 ## Cringe-Setup     11.790092  -1.086912 24.66710 0.0760466
 ## Payoff-Setup      7.906608  -2.867067 18.68028 0.1806133
 ## Cringe-Absurdity  7.747314  -2.766716 18.26134 0.1781887
 ## Payoff-Absurdity  3.863830  -3.933584 11.66124 0.4742692
 ## Payoff-Cringe    -3.883484 -14.657159  6.89019 0.7054376
```

```
pre_wps_anova <- aov(words_per_second ~ tension_cat, data = pre_2000)
 summary(pre_wps_anova)

##            Df Sum Sq Mean Sq F value Pr(>F)
 ## tension_cat  3  1.562  0.5207   0.991  0.433
 ## Residuals   11  5.782  0.5256

TukeyHSD(pre_wps_anova)

##  Tukey multiple comparisons of means
 ##      95% family-wise confidence level
 ##
 ## Fit: aov(formula = words_per_second ~ tension_cat, data = pre_2000)
 ##
 ## $tension_cat
 ##                          diff      lwr       upr     p adj
 ## Absurdity-Setup  -0.66504757 -2.446584 1.1164888 0.6834359
 ## Cringe-Setup     -0.95281908 -3.134747 1.2291084 0.5733005
 ## Payoff-Setup     -1.00269142 -2.828223 0.8228401 0.3910925
 ## Cringe-Absurdity -0.28777151 -2.069308 1.4937648 0.9605939
 ## Payoff-Absurdity -0.33764385 -1.658867 0.9835789 0.8666713
 ## Payoff-Cringe    -0.04987234 -1.875404 1.7756592 0.9997878

# Test 02 - The Modern Cinema
 # CPM & WPS
 print("UNDERSTANDING THE MODERN CINEMA")

## [1] "UNDERSTANDING THE MODERN CINEMA"

post_cpm_anova <- aov(cuts_per_minute ~ tension_cat, data = post_2000)
 summary(post_cpm_anova)

##            Df Sum Sq Mean Sq F value Pr(>F)
 ## tension_cat  3  162.3   54.09   0.494   0.69
 ## Residuals   26 2849.4  109.59

TukeyHSD(post_cpm_anova)

##  Tukey multiple comparisons of means
 ##      95% family-wise confidence level
 ##
 ## Fit: aov(formula = cuts_per_minute ~ tension_cat, data = post_2000)
 ##
```

```
## $tension_cat
##                          diff      lwr      upr      p adj
## Absurdity-Setup   6.586340 -13.720806 26.89349 0.8101646
## Cringe-Setup      4.616704 -13.921099 23.15451 0.9025460
## Payoff-Setup      7.976237 -11.169523 27.12200 0.6669842
## Cringe-Absurdity -1.969636 -16.328957 12.38968 0.9814386
## Payoff-Absurdity  1.389897 -13.746156 16.52595 0.9942485
## Payoff-Cringe     3.359534  -9.304197 16.02326 0.8850804
```

```
post_wps_anova <- aov(words_per_second ~ tension_cat, data = post_2000)
summary(post_wps_anova)
```

```
##             Df Sum Sq Mean Sq F value Pr(>F)
## tension_cat  3  0.483  0.1609   0.554   0.65
## Residuals   26  7.546  0.2902
```

```
TukeyHSD(post_wps_anova)
```

```
##  Tukey multiple comparisons of means
##    95% family-wise confidence level
##
## Fit: aov(formula = words_per_second ~ tension_cat, data = post_2000)
##
## $tension_cat
##                          diff      lwr      upr     p adj
## Absurdity-Setup   0.184518021 -0.8605465 1.2295825 0.9618901
## Cringe-Setup     -0.003098564 -0.9571076 0.9509105 0.9999997
## Payoff-Setup     -0.178893000 -1.1641893 0.8064033 0.9587851
## Cringe-Absurdity -0.187616584 -0.9265888 0.5513556 0.8974897
## Payoff-Absurdity -0.363411020 -1.1423561 0.4155341 0.5834453
## Payoff-Cringe    -0.175794436 -0.8275067 0.4759178 0.8800384
```

Probability prediction phase -

```
library(MASS) # Essential for prediction
```

```
##
## Attaching package: 'MASS'

## The following object is masked from 'package:dplyr':
##
##     select
```

```
# To ensure the order of the tension category
hcd$tension_cat <- factor(hcd$tension_cat,
                          levels = c("Setup", "Absurdity", "Cringe", "Payoff"),
                          ordered = TRUE)
# Model building - Edits (CPM) and Words (WPS) predicts the Tension category
prediction_model <- polr(tension_cat ~ cuts_per_minute + words_per_second,
                         data = hcd,
                         Hess = TRUE)
summary(prediction_model)
```

```
## Call:
## polr(formula = tension_cat ~ cuts_per_minute + words_per_second,
##      data = hcd, Hess = TRUE)
##
## Coefficients:
##                    Value Std. Error t value
## cuts_per_minute   0.04446    0.03256   1.365
## words_per_second -0.84310    0.49174  -1.715
##
## Intercepts:
##                  Value   Std. Error t value
## Setup|Absurdity  -2.8211  1.0655    -2.6478
## Absurdity|Cringe -1.0970  0.9727    -1.1278
## Cringe|Payoff     0.3193  0.9574 0.3335
##
## Residual Deviance: 113.0577
## AIC: 123.0577
```

```
# Odds-Ratio for easy interpretation
odds_ratio <- exp(coef(prediction_model))
print(odds_ratio)
```

```
## cuts_per_minute words_per_second
##       1.0454629       0.4303757
```

Framing a technical blueprint table -

```
library(dplyr)
library(knitr) # For organized table
```

```
# Calculating the average CPM and WPS for every era and tension category combination
 directive_data <- hcd %>%
   group_by(era, tension_cat) %>%
   summarise(
          Mean_CPM = round(mean(cuts_per_minute, na.rm = TRUE), 1),
          Mean_WPS = round(mean(words_per_second, na.rm = TRUE), 2),
          .groups = "drop"
   )

 # Directives based on Odds-Ratios
 technical_directive_table <- directive_data %>%
 mutate(
          Narrative_Goal = case_when(
          tension_cat == "Setup"          ~ "Contextual Foundation",
          tension_cat == "Absurdity" ~ "Rising Complexity",
          tension_cat == "Cringe"         ~ "Social Friction",
          tension_cat == "Payoff"         ~ "Rhythmic Climax"
          ),
          Actionable_Directive = case_when(
          tension_cat == "Setup"          ~ "Prioritize verbal clarity; keep visual rhythm stable.",
          tension_cat == "Absurdity" ~ "Increase technical energy; accelerate CPM by ~50%.",
          tension_cat == "Cringe"         ~ "Silence is key. Drop WPS to amplify audience
discomfort.",
          tension_cat == "Payoff"         ~ "Peak Velocity. Maximize CPM; let visual cuts drive
humor."
          )
   )

 # Displaying the Final Table using Kable (For report-style table)
 kable(technical_directive_table,
          col.names = c("Era", "Category", "Target CPM", "Target WPS", "Technical
Directive", "Narrative Goal"),
          caption = "Technical Directive Table for Cinematic Tension Engineering",
 align = c("l", "l", "l", "c", "c", "l")) # l = left-align, c = center-align
```

Output: *Figure 6.2.5: Technical Directive table*

MACHINE LEARNING -

Importing the necessary libraries -

```
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_squared_error,mean_absolute_error
```

Loading the data set -

```
hcd = pd.read_csv('/content/Project_Master_Data.csv')
```

Adding new metrics -

```
hcd['CPM'] = hcd['Total_Cuts'] / (hcd['Duration_Sec'] / 60)
hcd['WPS'] = hcd['Word_Count'] / hcd['Duration_Sec']
```

Dividing the data -

```
# Precautionary step for handling the NAN values (if any)
hcd.replace([np.inf,-np.inf],np.nan,inplace=True)
hcd.dropna(subset=['CPM','WPS','Tension_Score'],inplace=True)

X = hcd[['CPM','WPS']] # Input
y = hcd['Tension_Score'] # Output
```

Train-test splitting phase -

```
X_train, X_test, y_train, y_test =
train_test_split(X,y,test_size=0.2,random_state=42)
```

Model training phase -

```
basic_model = RandomForestRegressor(n_estimators=100,random_state=42)
basic_model.fit(X_train,y_train)
```

```
▾ RandomForestRegressor
RandomForestRegressor(random_state=42)
```

Model prediction and evaluation -

```
y_pred = basic_model.predict(X_test) # Output is predicted using the
test input
mse = mean_squared_error(y_test,y_pred) # Mean Squared Error
mae = mean_absolute_error(y_test,y_pred) # Mean Absolute Error
print(f"Model Evaluation shows MSE at {mse:.2f} and MAE at {mae:.2f}")
```

Model Evaluation shows MSE at 6.42 and MAE at 2.09

Prediction architecture creation -

```
def tension_prediction(cpm,wps):
  input_data = pd.DataFrame({'CPM':[cpm],'WPS':[wps]}) # Exclusively
for input values
  score_predicted = basic_model.predict(input_data)[0]
  rounded_score = round(score_predicted)
  # Mapping the rounded predicted score to a tension category
  if 1 <= rounded_score <= 3:
        category = "Casual/Set-up - Characters are just talking,
reflecting low energy, providing information"
  elif 4 <= rounded_score <= 6:
        category = "Rising Absurdity - Some sort of misunderstanding is
growing"
  elif 7 <= rounded_score <= 8:
        category = "Cringe/Chaos - Extreme social awkwardness or
physical slapstick is happening"
  elif 9 <= rounded_score <= 10:
        category = "The Big Payoff - The punchline hits or the chaotic
situation explodes"
  else:
        category = "Out of standard bounds"
  # Displaying the results
  print("\n--- Prediction Results ---")
  print(f"Input Pacing Setup -> CPM: {cpm}, WPS: {wps}")
  print(f"Predicted Raw Tension Score: {score_predicted}")
  print(f"Tension Category: {category}")
```

Generating the prediction output -

```
tension_prediction(45.0,4.0)
```

Output: *Figure 6.3.2: Predicting the tension category*